Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Differential Equations

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $\dfrac{dy}{dx} = 3y$. **[3]**

2. Solve $\dfrac{dy}{dx} = 2x$. **[2]**

3. Solve $\dfrac{dy}{dx} = \frac{y}{x}$. **[3]**

4. Solve $\dfrac{dy}{dx} + 2y = 0$ given $y(0) = 5$. **[4]**

5. Find the general solution of $\dfrac{dy}{dx} = y\cos x$. **[4]**

6. Solve $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = 0$. **[4]**

7. Solve $\dfrac{d^2y}{dx^2} + 4y = 0$. **[4]**

8. Solve $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = 0$. **[4]**

9. Find the complementary function of $\dfrac{d^2y}{dx^2} + 5\dfrac{dy}{dx} + 6y = e^x$. **[3]**

10. Find a particular integral for $\dfrac{d^2y}{dx^2} + y = x$. **[3]**

11. Solve $\dfrac{dy}{dx} + y = e^{-x}$. **[4]**

12. State the form of the CF when the auxiliary equation has equal roots. **[2]**

13. State the form of the CF when the auxiliary equation has complex roots $\alpha \pm i\beta$. **[2]**

14. Solve $\dfrac{dy}{dx} = x + y$ using an integrating factor or substitution. **[5]**

15. Verify that $y = Ae^{2x} + Be^{-x}$ satisfies $\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx} - 2y = 0$. **[3]**

Section B – Hard (Q16–25)

16. Solve $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = e^{3x}$. **[6]**

17. Solve $\dfrac{d^2y}{dx^2} + y = \sin x$. **[5]**

18. Solve $\dfrac{dy}{dx} + \frac{2}{x}y = x$ given $y(1) = 2$. **[5]**

19. Find the general solution of $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + y = xe^{x}$. **[6]**

20. Solve $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 5y = 0$ given $y(0) = 1$, $y'(0) = 0$. **[5]**

21. Use the substitution $v = \dfrac{dy}{dx}$ to solve $y\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 = 0$. **[5]**

22. Solve $\dfrac{dy}{dx} = \frac{x+y}{x-y}$. **[5]**

23. Find the particular solution of $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = 4$ with $y(0) = 1$, $y'(0) = 0$. **[6]**

24. Solve $\dfrac{d^2y}{dx^2} + \omega^2 y = 0$ and interpret as SHM. **[4]**

25. Using an integrating factor, solve $x\dfrac{dy}{dx} - 2y = x^3$. **[5]**

End of Worksheet • *Show all working clearly.*